



JOHNS HOPKINS

WHITING SCHOOL
of ENGINEERING

FPGA Design Using VHDL

Module 4B: Simulator Operation

Simulation Statement

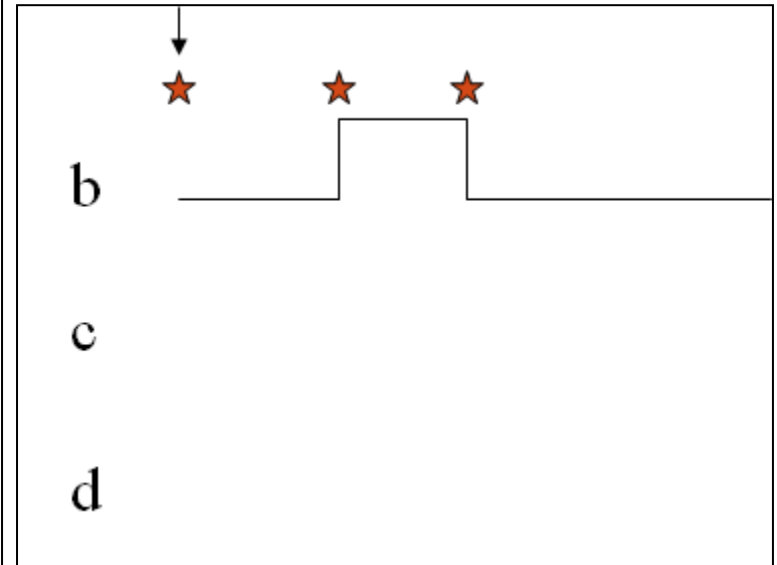
- Statements in a process are executed in order in a simulator
- Each statement is executed every **delta cycle**
 - All statements are executed in zero **simulation time**
- All **signals** are **updated** at the **end** of the process
 - In simulation
- All variables are updated **immediately**
 - In simulation

Delta Time

- Infinitely small (δ) advance in time
- The current simulation time TC does not advance as δ advances
 - $TC + \delta = TC$
 - $TC + 100 \delta = TC$
- Infinite number of delta time steps can occur during current simulation time TC
- Simulator places statements in order
- Statements in process updated at end of process

Process Example 1

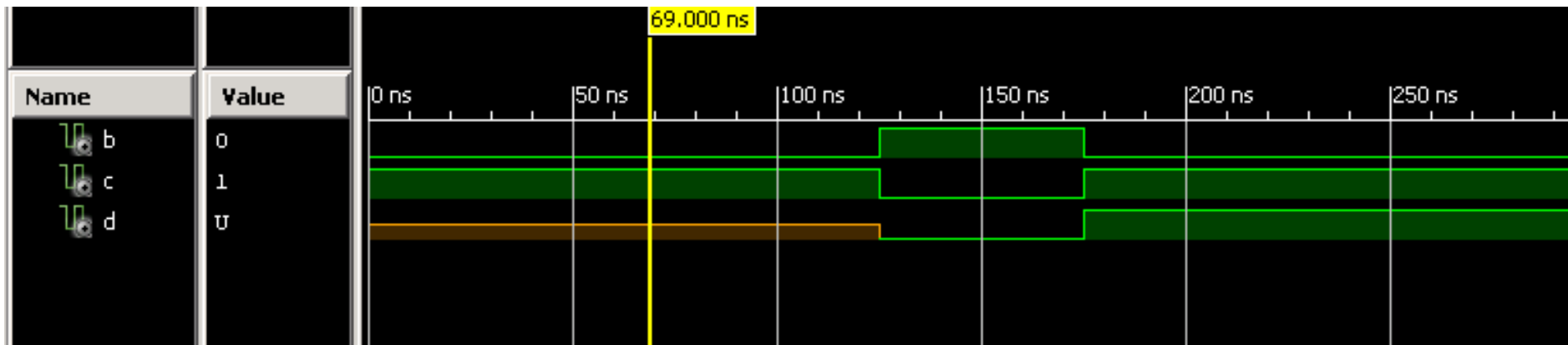
```
example : process(b)
begin
    c <= not b;
    d <= not c;
    c <= b;
    c <= not b;
end process example;
```



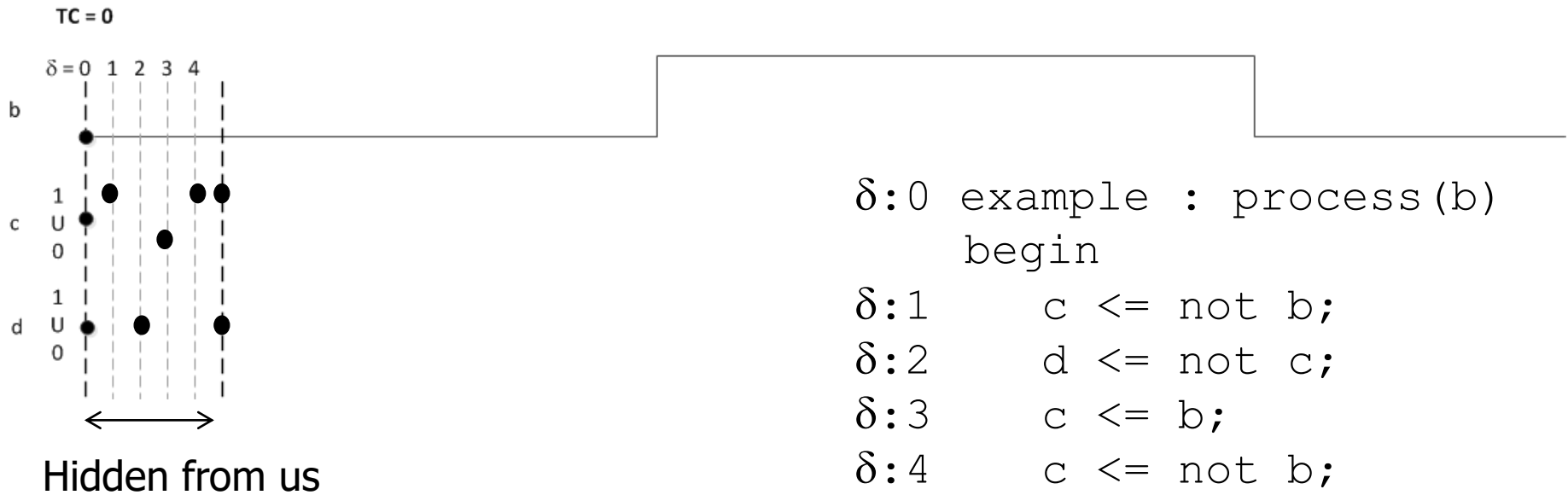
- All statements executed in zero simulation time
- All signals are updated at the end of the process

Process Example 2

```
example : process(b)
begin
    c <= not b;
    d <= not c;
    c <= b;
    c <= not b;
end process example;
```



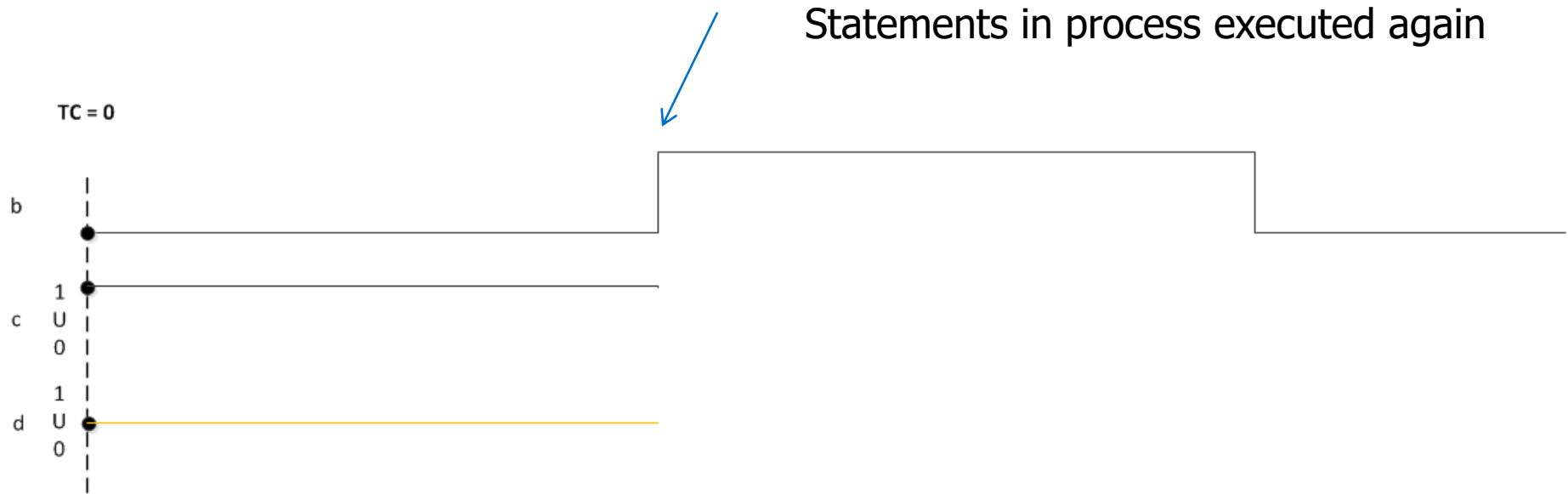
Process Example 3



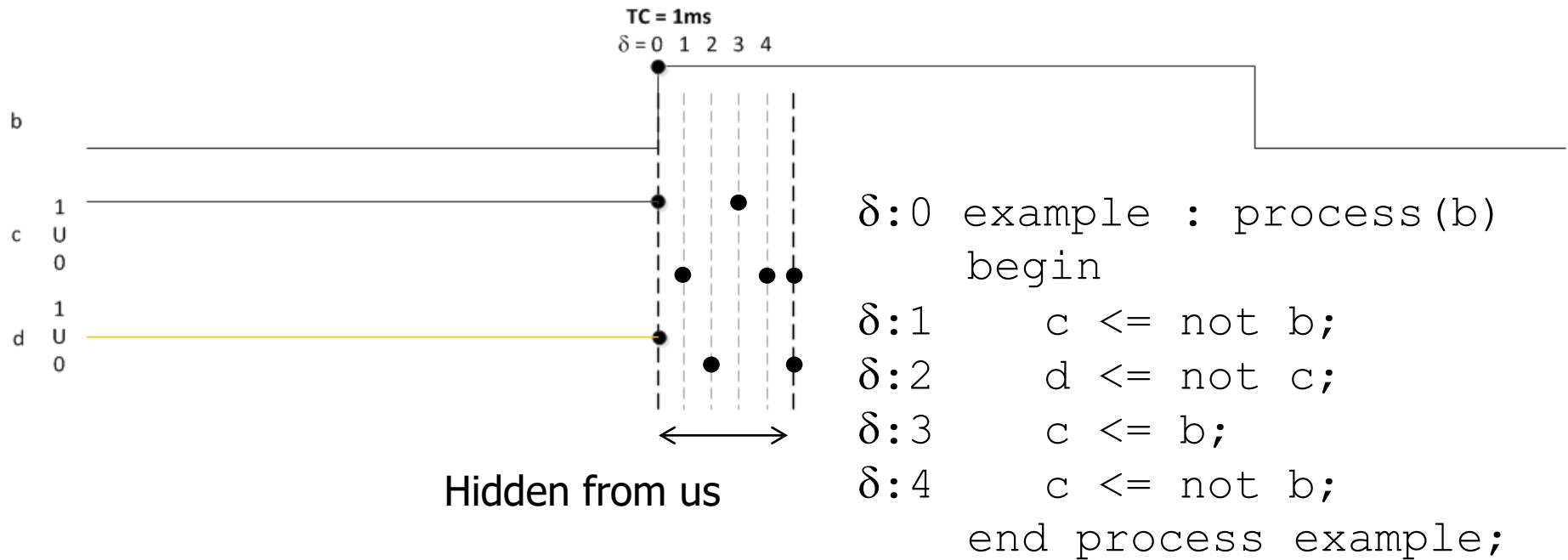
```

 $\delta:0$  example : process(b)
      begin
 $\delta:1$       c <= not b;
 $\delta:2$       d <= not c;
 $\delta:3$       c <= b;
 $\delta:4$       c <= not b;
      end process example;
  
```

Process Example 4



Process Example 5



Process Example 6

Statements in process executed again

